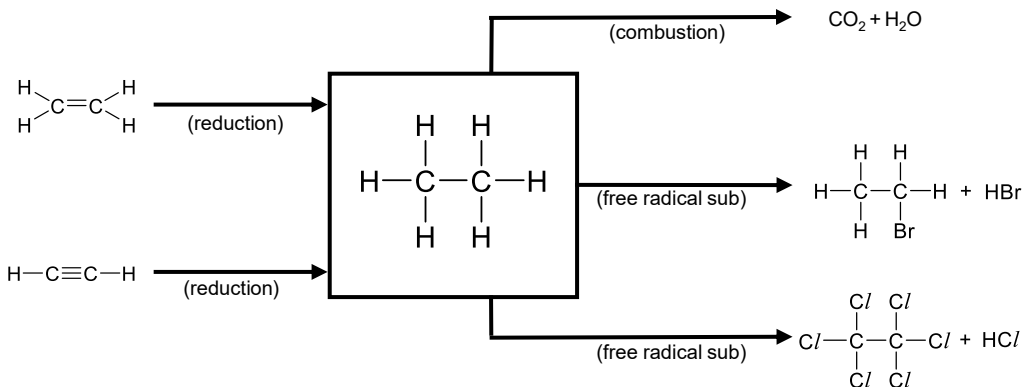
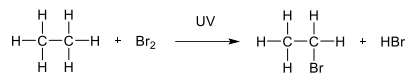


Reactions



Mechanism

Describe the mechanism of the following reaction:



Name of mechanism:

Initiation:

Propagation:

Termination:

Notes

- must draw fish-hook arrow in initiation step
- must label UV
- not required to draw arrow in propagation step, but need to know how it works
- can also draw mechanism using skeletal structures
- count the number of C atoms in the product of termination check to double-check

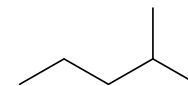
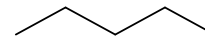
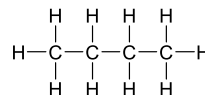
Explaining general reactivity

Explain why alkanes are generally unreactive.

- 1 They are _____ and therefore there is no _____ to attract _____ and no _____ to attract _____
- 2 They contain strong _____ which require large amount of energy to break

Deducing no. of isomers from FRS

Deduce the number of isomers (excluding stereoisomerism) that can be formed from the mono-bromination of the following molecules.

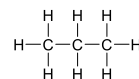


no. of types of H: _____
no. of products: _____

no. of types of H: _____
no. of products: _____

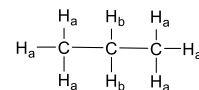
no. of types of H: _____
no. of products: _____

Factors affecting ratio of isomers from FRS



When propane undergoes mono-bromination, two isomers are formed: 1-bromopropane and 2-bromopropane. However, they are not formed in a 1:1 ratio. What are the factors which affects the ratio of the isomers formed?

1 Probability factor

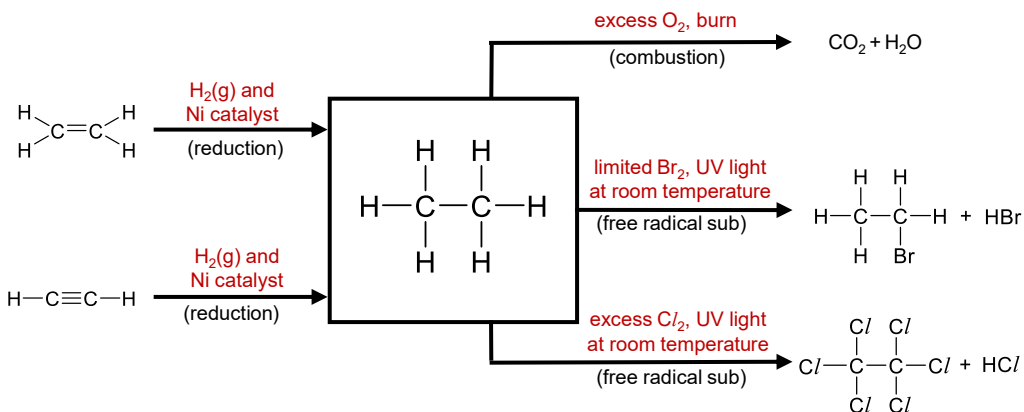


- there are two types of H in propane: H_a and H_b in the ratio of _____
- substituting any of the _____ with Br will result in the formation of 1-bromopropane.
- substituting any of the _____ with Br will result in the formation of 2-bromopropane.
- therefore, by probability factor, the reaction will favor the formation of _____

2 Reactivity factor

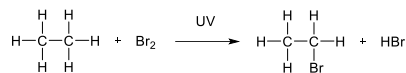
- formation of 1-bromopropane proceeds through the formation of the radical: _____
- formation of 2-bromopropane proceeds through the formation of the radical: _____
- the _____ radical is more stable compared to the _____ radical, because the radical C is bonded to an additional _____ group which stabilises the _____ radical C
- therefore, by reactivity factor, the reaction will favor the formation of radical: _____ which further reacts to form _____

Reactions

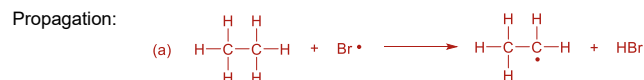


Mechanism

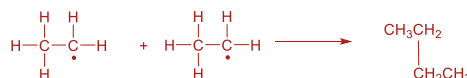
Describe the mechanism of the following reaction:



Name of mechanism: **Free radical substitution**



then (a), (b), (a), (b)...



Notes

- must draw fish-hook arrow in initiation step
- must label UV
- not required to draw arrow in propagation step, but need to know how it works
- can also draw mechanism using skeletal structures
- count the number of C atoms in the product of termination check to double-check

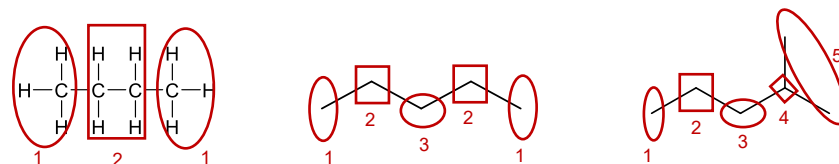
Explaining general reactivity

Explain why alkanes are generally unreactive.

- 1 They are **non-polar** and therefore there is no **electron rich sites** to attract **electrophiles** and no **electron deficient sites** to attract **nucleophiles**
- 2 They contain strong **C-C** and **C-H** bonds which require large amount of energy to break

Deducing no. of isomers from FRS

Deduce the number of isomers (excluding stereoisomerism) that can be formed from the mono-bromination of the following molecules.

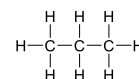


no. of types of H: 2
no. of products: 2

no. of types of H: 3
no. of products: 3

no. of types of H: 5
no. of products: 5

Factors affecting ratio of isomers from FRS



When propane undergoes mono-bromination, two isomers are formed: 1-bromopropane and 2-bromopropane. However, they are not formed in a 1:1 ratio. What are the factors which affects the ratio of the isomers formed?

1 Probability factor

- $\text{H}_a-\text{C}-\text{C}-\text{C}-\text{H}_a$
- there are two types of H in propane: H_a and H_b in the ratio of 6 : 2
 - substituting any of the H_a with Br will result in the formation of 1-bromopropane.
 - substituting any of the H_b with Br will result in the formation of 2-bromopropane.
 - therefore, by probability factor, the reaction will favor the formation of 1-bromopropane

2 Reactivity factor

- formation of 1-bromopropane proceeds through the formation of the radical: $\text{H}-\text{C}-\text{C}-\text{C}-\text{H}$
- formation of 2-bromopropane proceeds through the formation of the radical: $\text{H}-\text{C}-\text{C}-\text{C}-\text{H}$
- the secondary radical is more stable compared to the primary radical, because the radical C is bonded to an additional electron-donating alkyl group which stabilises the electron-deficient radical C
- therefore, by reactivity factor, the reaction will favor the formation of radical: secondary which further reacts to form 2-bromopropane